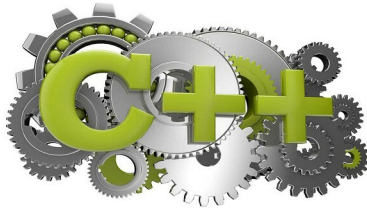


Regular Expressions in Programming Languages: The Story of C++

In this issue of *OSFY*, we present the third article on regular expressions in programming languages. The earlier articles covered the use of regular expressions in general, in Python and then in Perl. Read on to discover the intricacies of regular expressions in C++.



Interpreted languages often have weakly typed variables which don't require prior declaration. The additional benefit of weakly typed variables is that they can be used to hold different types of data. For example, the same variable can hold an integer, a character, or a string. Due to these qualities, scripts written in such languages are often very compact. But this is not the case with compiled languages, for which you need a lot of initialisation; and with strongly typed variables, the code is often longer. Even if the regular expression syntax for interpreted and compiled languages is the same, how they are used in real programs is different. So, I believe it is time to discuss regular expressions in compiled languages. In this article, I will discuss the regular expression syntax of C++.

Standards of C++

People often fail to notice the fact that programming languages like C and C++ have different standards. This is quite unlike languages like Perl and Python, for which the use of regular expressions is highly warranted due to the very nature of these programming languages (they are scripting languages widely used for text processing and Web application development).

For a language like C++, heavily used for high-performance computing applications, system programming, embedded system development, etc, many felt that the inclusion of regular expressions was unnecessary. Many of the initial standards of C++ didn't have a natural way for handling regular expressions. I will briefly discuss the different standards of C++ and which among them has support for regular expressions.

C++ was invented by Bjarne Stroustrup in 1979 and was initially known as 'C with Classes' and later renamed C++ in

1983. A book titled 'The C++ Programming Language', first published in 1985 by Stroustrup himself, and its subsequent editions became the *de facto* standard for C++ until 1998, when the language was standardised by the International Standards Organization (ISO) and the International Electrotechnical Commission (IEC) as ISO/IEC 14882:1998, informally called C++98. The next three standards of C++ are informally called C++03, C++11 and C++14. Hopefully, by the time this article gets published, the latest standard of C++, informally called C++17, would have been released and will have some major changes to C++. After this, the next big changes in C++ will take place with a newer standard, informally known as C++20, which is set to be released in 2020.

The first three standards of C++, namely the *de facto* standard of C++, C++98 and C++03, do not have any inbuilt mechanism for handling regular expressions. Things changed with C++11 when native support for regular expressions was added with the help of a new header file called `<regex>`. In fact, the support for regular expressions was one of the most important changes brought in by this standard. C++14 also has provision for native support of regular expressions, and it is highly unlikely that C++17 or any future standards of C++ will quash the support for handling regular expressions. One problem we might face in this regard is that the academic community in India mostly revolves around the C++98 standard, which doesn't support regular expressions. But this is just a personal opinion and I don't have any documented evidence to prove my statement.

The C++11 standard

Unlike C++03 and C++14, for which the changes were minimal, C++11 was a major revision of C++. GCC 5 fully